

Study of pn Junction Solar Cell Characteristics and Performance Analysis

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Abstract—Solar energy is one of the most promising renewable energy sources available today. The pn junction solar cell converts solar radiation directly into electrical energy through the photovoltaic effect. This paper presents a detailed study of the working principle of a pn junction solar cell, its current-voltage (I-V) characteristics, important parameters such as short circuit current, open circuit voltage, fill factor, and efficiency. The suitability of semiconductor materials based on energy band gap for solar cell applications is also discussed.

Index Terms—Solar cell, pn junction, photovoltaic effect, I-V characteristics, fill factor, efficiency.

I. INTRODUCTION

The increasing demand for energy and the depletion of fossil fuels have led to a greater focus on renewable energy resources. Solar energy is abundant, clean, and sustainable. A solar cell is a semiconductor device that converts sunlight directly into electrical energy using the photovoltaic effect.

The most commonly used solar cell is the silicon pn junction solar cell due to its availability, stability, and suitable energy band gap. Understanding the electrical characteristics of a solar cell is essential for improving its efficiency and performance.

II. CONSTRUCTION OF PN JUNCTION SOLAR CELL

A pn junction solar cell consists of a thin p-type and n-type semiconductor layer formed by doping pure silicon with trivalent and pentavalent impurities respectively.

- The top surface is coated with an anti-reflection coating.
- A metallic grid is used as the front contact.
- The back surface has a full metal contact.
- The junction region creates an internal electric field.

This built-in electric field plays a crucial role in separating the photo-generated electron-hole pairs.

III. WORKING PRINCIPLE

When sunlight falls on the solar cell, photons with energy greater than the band gap of the semiconductor are absorbed. This absorption creates electron-hole pairs.

Due to the internal electric field present at the pn junction:

- Electrons move towards the n-region.
- Holes move towards the p-region.

This movement of charge carriers produces a photocurrent. When an external load is connected, current flows through the circuit.

IV. I-V CHARACTERISTICS OF SOLAR CELL

The current-voltage (I-V) characteristics of a solar cell describe the relationship between the output current and voltage under illumination.

A. Important Parameters

- Short Circuit Current (I_{sc})
- Open Circuit Voltage (V_{oc})
- Maximum Power Point (MPP)
- Fill Factor (FF)

B. I-V Characteristics Graph

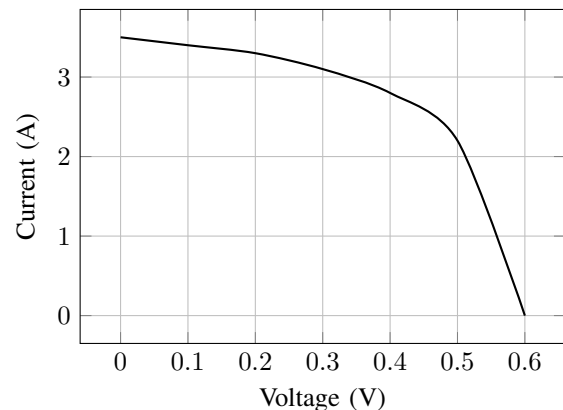


Fig. 1: I-V Characteristics of pn Junction Solar Cell

V. SHORT CIRCUIT CURRENT AND OPEN CIRCUIT VOLTAGE

A. Short Circuit Current

Short circuit current (I_{sc}) is the maximum current obtained from a solar cell when the output terminals are shorted. It depends on:

- Incident light intensity
- Area of the cell
- Material properties

B. Open Circuit Voltage

Open circuit voltage (V_{oc}) is the maximum voltage obtained when no current flows through the circuit. It mainly depends on:

- Band gap of the semiconductor of charge separation

VI. EFFICIENCY CALCULATION

The efficiency of a solar cell is given by:

$$\eta = \frac{P_{out}}{P_{in}}$$

Where:

$$P_{out} = V_{oc} \times I_{sc} \times FF$$

Given:

$$I_{sc} = 35 \text{ mA/cm}^2$$

$$V_{oc} = 0.60 \text{ V}$$

$$FF = 0.80$$

$$P_{in} = 100 \text{ mW/cm}^2$$

$$P_{out} = 0.60 \times 35 \times 0.80 = 16.8 \text{ mW/cm}^2$$

$$\eta = \frac{16.8}{100} = 16.8\%$$

VII. ENERGY BAND GAP AND MATERIAL SUITABILITY

The energy band gap of a semiconductor plays a crucial role in solar cell applications.

If the band gap is too large (e.g., 3 eV):

- Most of the solar spectrum is not absorbed
- Photocurrent is very low
- Overall efficiency is poor

Therefore, a material with a band gap around 1.1–1.6 eV is more suitable for solar cells.

VIII. CONCLUSION

The pn junction solar cell is an effective device for converting solar energy into electrical energy. The performance of the solar cell depends on its I–V characteristics, material properties, and operating conditions. Silicon remains the most widely used material due to its suitable band gap and stability.

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